

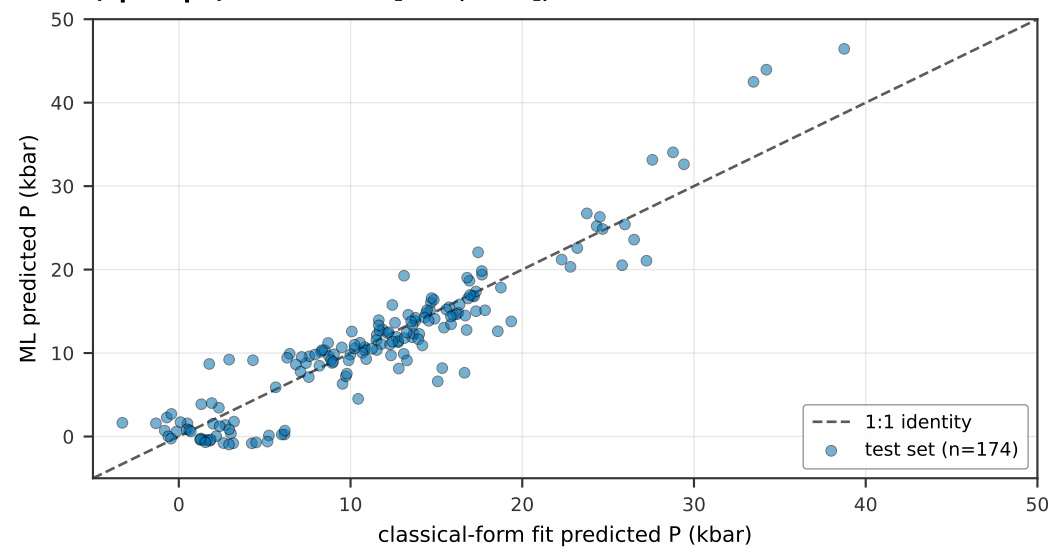
Core_16. Classical-equivalence regression: per-track winner ML prediction regressed on Putirka-form classical features.

Rows: opx-liq, opx-only, cpx-liq, cpx-only. Cols: P 1:1, T 1:1, ML-minus-classical residual.

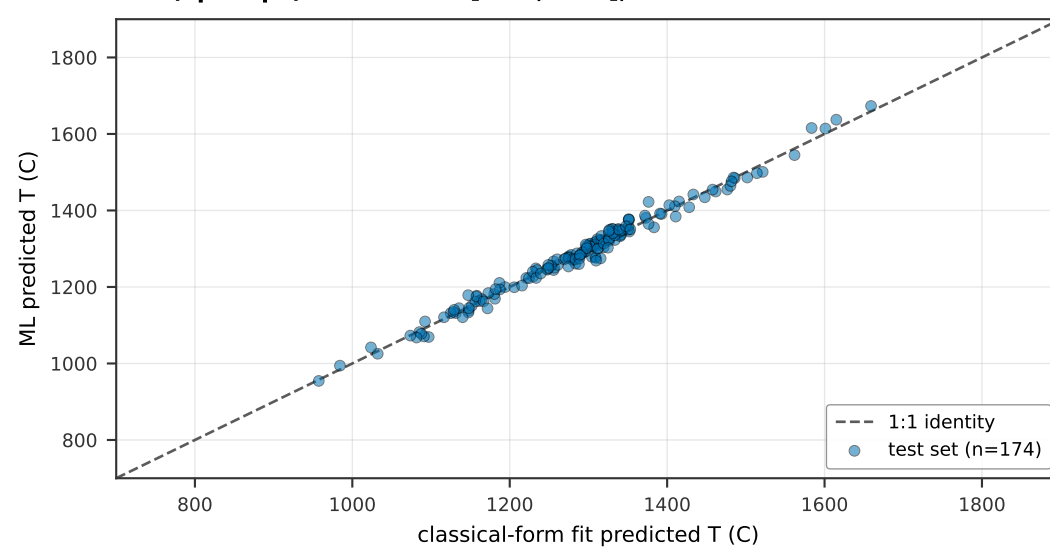
Axes fixed: P in (-5.0, 50.0) kbar, T in (700.0, 1900.0) C, DeltaP in (-20.0, 20.0) kbar, DeltaT in (-300.0, 300.0) C.

Box = +/- 2 sigma_conformal (sigma_P = 10.0 kbar, sigma_T = 92.6 C from nb07).

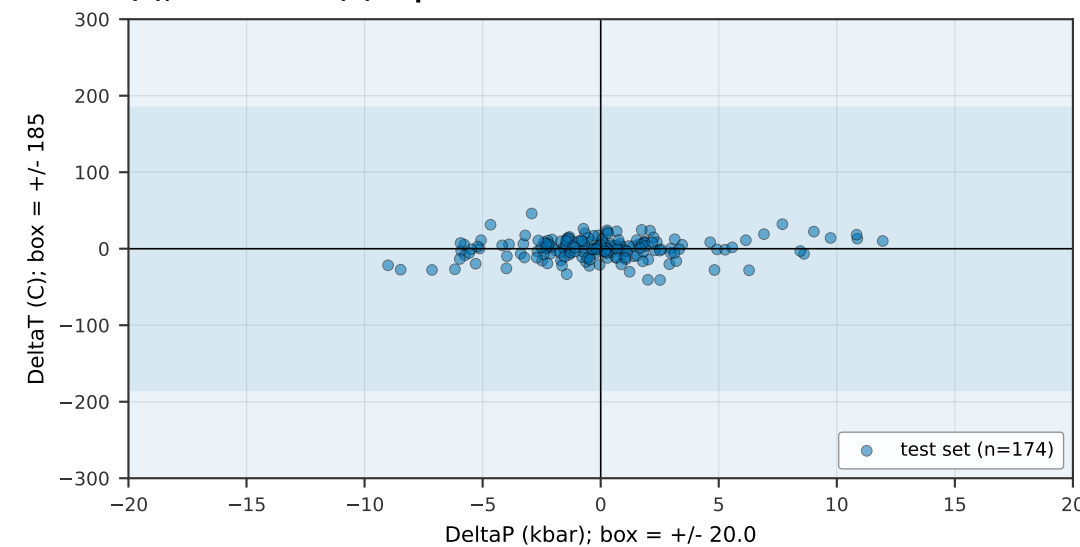
(A1) P: opx_liq winner MLP/raw vs classical Putirka 29a (opx-liq P). $R^2 = 0.86$ [0.79, 0.89], n = 174



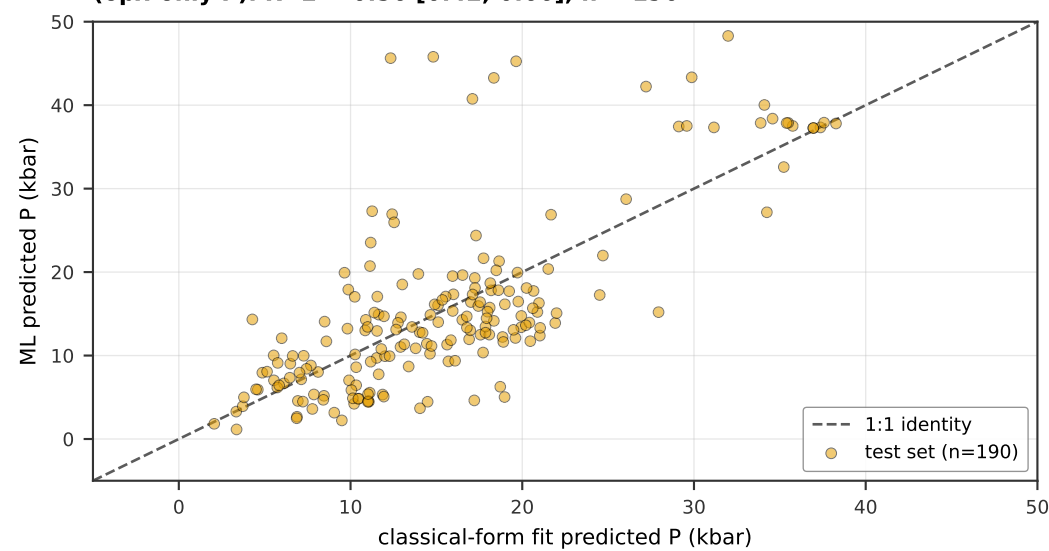
(A2) T: opx_liq winner ElasticNet/raw vs classical Putirka 28a (opx-liq T). $R^2 = 0.99$ [0.98, 0.99], n = 174



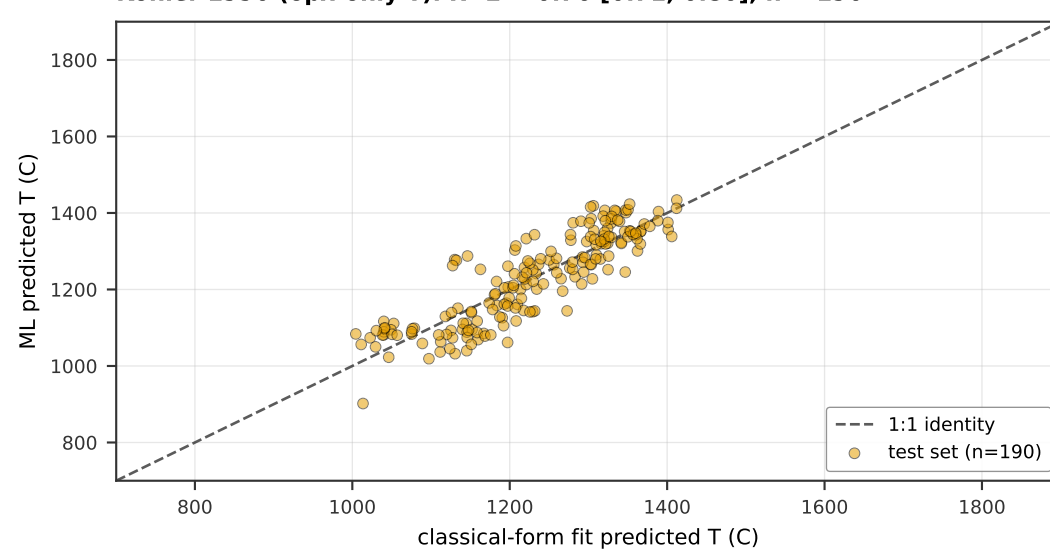
(A3) ML minus classical-fit residuals, opx_liq: Putirka 28a (T), Putirka 29a (P). equilib. box 100.0%



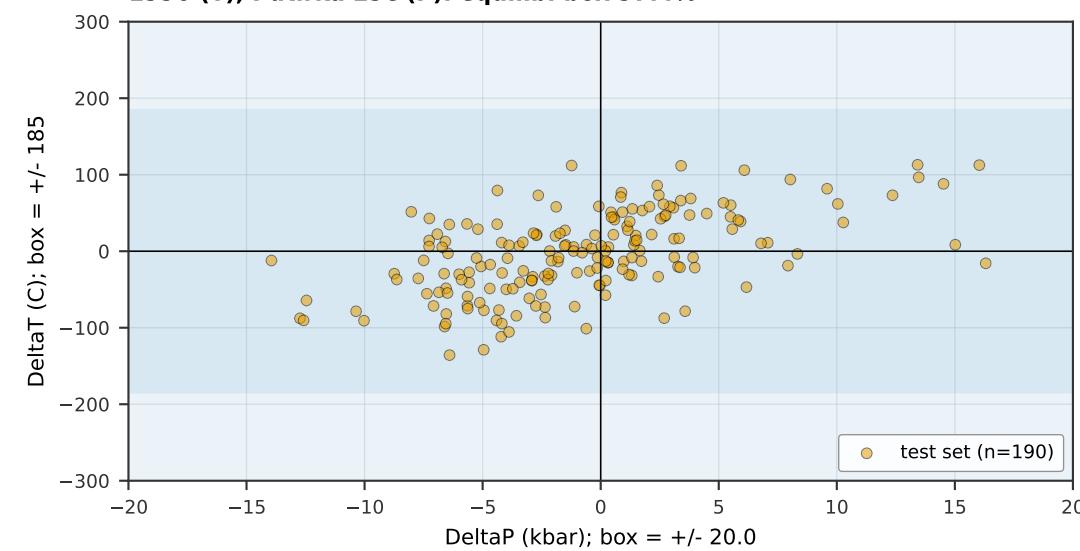
(B1) P: opx_only winner RF/pwlr vs classical Putirka 29c (opx-only P). $R^2 = 0.56$ [0.42, 0.66], n = 190



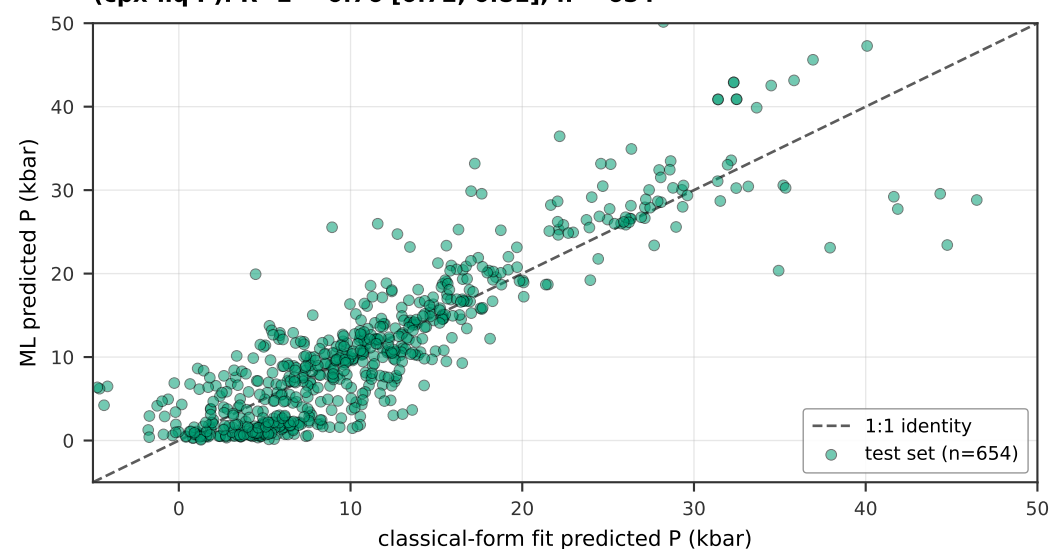
(B2) T: opx_only winner LightGBM/alr vs classical Brey-Kohler 1990 (opx-only T). $R^2 = 0.76$ [0.71, 0.80], n = 190



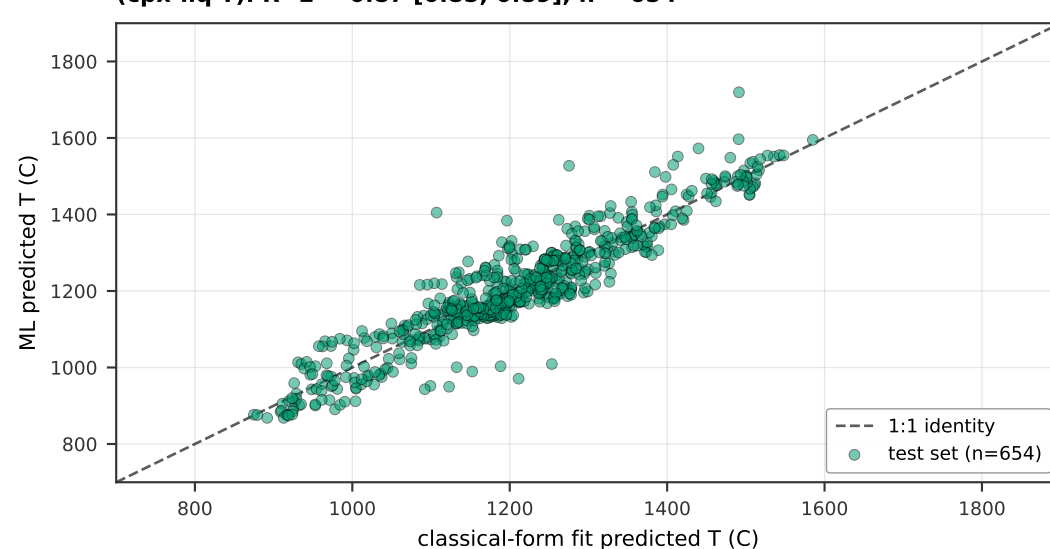
(B3) ML minus classical-fit residuals, opx_only: Brey-Kohler 1990 (T), Putirka 29c (P). equilib. box 97.4%



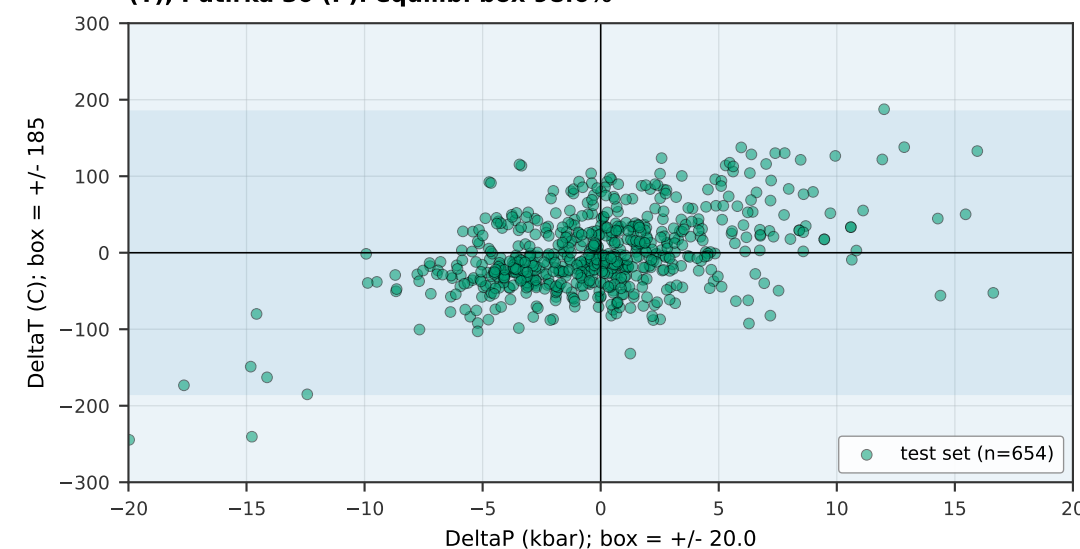
(C1) P: cpx_liq winner LightGBM/pwlr vs classical Putirka 30 (cpx-liq P). $R^2 = 0.76$ [0.72, 0.81], n = 654



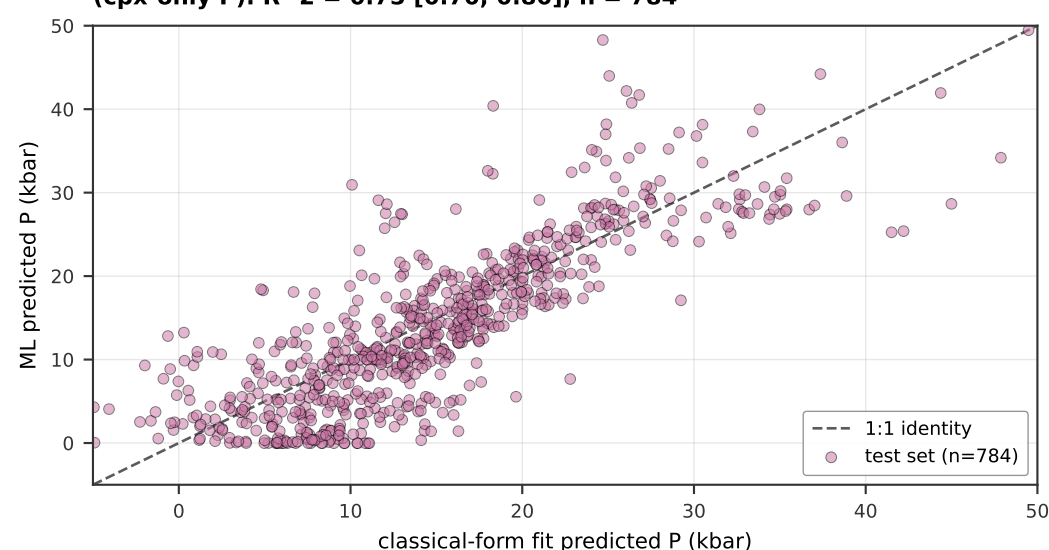
(C2) T: cpx_liq winner TabPFN/raw vs classical Putirka 33 (cpx-liq T). $R^2 = 0.87$ [0.85, 0.89], n = 654



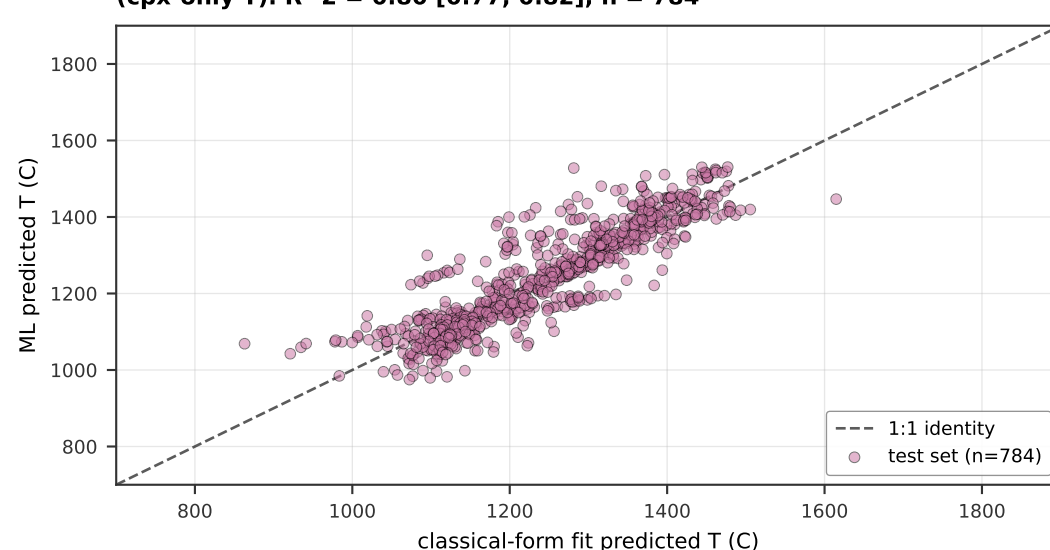
(C3) ML minus classical-fit residuals, cpx_liq: Putirka 33 (T), Putirka 30 (P). equilib. box 98.6%



(D1) P: cpx_only winner TabPFN/raw vs classical Putirka 32a (cpx-only P). $R^2 = 0.75$ [0.70, 0.80], n = 784



(D2) T: cpx_only winner ERT/pwlr vs classical Putirka 32d (cpx-only T). $R^2 = 0.80$ [0.77, 0.82], n = 784



(D3) ML minus classical-fit residuals, cpx_only: Putirka 32d (T), Putirka 32a (P). equilib. box 96.2%

